Document Identifier: **PMC8544130**

SciScore Report

Below you will find your SciScore report containing three tables. Your score is calculated based on adherence to scientific rigor criteria (Table 1) and identification of key biological resources (Table 2). Table 3 contains statistical tests and oligonucleotides but is not scored. If SciScore makes any mistakes, please contact us to help us learn and improve.

Table 1: Rigor Adherence Table

Ethics
IRB: Methods Ethics Statement This study was reviewed and approved by the ethical committee of the Islamia University of Bahawalpur, Bahawalpur, Pakistan.
Inclusion and Exclusion Criteria
not detected.
Attrition
not detected.
Sex as a biological variable
not detected.
Subject Demographics
Age: not detected.
Weight: not detected.
Randomization
not detected.
Blinding
not detected.
Power Analysis
not detected.
Replication
not required.

Table 2: Key Resources Table

Your Sentences	REAGENT or RESOURCE	SOURCE	IDENTIFIER
Software and Algorithms			
UALCAN Transcription expression level variations of TMED2 in different human cancers relative to normal controls were analyzed using UALCAN database (http:ualcan.path.uab.edu).	UALCAN		Suggestion: UALCAN (RRID:SCR_015827)(link)
GENT2 Database GENT2 (http:gent2.appex.kr) web server is developed for the multi-omics analysis of the cancer data acquired from Gene Expression Omnibus (GEO, https:www.ncbi.nlm.nih.govgeo) database.10 In our study, we utilized GENT2 for the validation of TMED2 expression using new independent cohorts of distinct cancer patients.	Gene Expression Omnibus		Suggestion: Gene Expression Omnibus (GEO) (RRID:SCR_005012)(link)
Data Mining Through Human Protein Atlas Database HPA (https:www.proteinatlas.org) is a database that aims to quantify and locate all the human proteins in cells, tissues, and organs using the immunohistochemistry (IHC) technique.11 In the current study, HPA was utilized to examine the proteomics expression level of TMED2 in normal tissues and distinct subtypes of cancerous tissues.	https:www.proteinatlas.org		
The OncoPrint option on the cBioPortal provides an overview of genetic mutations and CNVs in the gene(s) of interest above every sample bar.	cBioPortal		Suggestion: cBioPortal (RRID:SCR_014555)(link)
TMED2 Protein-Protein Interaction Network Development and Pathway Analysis A PPI network of TMED2-enriched genes was developed via STRING (https:string-db.org)14 and visualized through Cytoscape software.	STRING		
	Cytoscape		Suggestion: Cytoscape (RRID:SCR_003032)(link)
15 Then, the pathway analysis of the TMED2-enriched genes was carried out using DAVID (http:david.ncifcrf.govsummary.jsp)16 and a p value < 0.05 was considered as significant.	DAVID		

Other Entities Detected

Your Sentences	Recognized Entity
Statistical Tests	
9 In UALCAN, the transcription expression level of TMED2 was quantified as transcript per million (TPM) reads, and a Students t-test was employed for statistics purpose.	t-test
MEXPRESS Database MEXPRESS database was utilized to assess the Pearson correlation between TMED2 transcription expression and its promoter methylation levels in 309, 194, 542, 343, 414, and 477 samples of CESC, ESCA, HNSC, KIRC, LIHC, and LUAD, respectively.	Pearson correlation
TIMER applied the Spearman correlation method to infer the abundance of immune cells in association with gene expression using TCGA datasets.	Spearman
In distinct human cancer subtypes, the Spearman correlation between the TMED2 expression and CD8+ T immune cells was calculated in this study using TIMER.	Spearman correlation

SciScore is an automated tool that is designed to assist expert reviewers by finding and presenting formulaic information scattered throughout a paper in a standard, easy to digest format. ***SciScore is not a substitute for expert review.*** SciScore also checks for the presence and correctness of several unique identifiers, including RRIDs (research resource identifiers) in the manuscript, detects sentences that appear to be missing RRIDs, and can even suggest RRIDs under certain circumstances. **All RRID suggestions should be verified;** only the author can know whether the suggestions are correct.

For a full description of scored criteria and tips for improving your score, please see https://www.scicrunch.com/sciscorereport-faq